

# Membership Inference Attack Defense Report

## 1. Introduction

### 1.1 Background

Machine learning models can inadvertently memorize training data, making them vulnerable to membership inference attacks. An attacker with access to the model's predictions can determine whether a specific data point was part of the training set, leading to privacy breaches.

### 1.2 Objectives

- Quantify the vulnerability of a baseline CNN to membership inference attacks
- Implement and evaluate five defense strategies
- Analyze the privacy-utility tradeoff
- Provide actionable recommendations

### 1.3 Dataset and Model

- Dataset: CIFAR-10 (50,000 training images, 10,000 test images)
- Model: SimpleCNN with 3 convolutional blocks and 2 fully connected layers
- Attack Setup:
  - Members: 45,000 training samples
  - Non-members: 10,000 test samples
  - Attack Signal: Maximum softmax confidence

## 2. Threat Model: Membership Inference Attack

### 2.1 Attack Mechanism

The membership inference attack exploits the observation that models tend to be more confident on training data (members) than on unseen data (non-members).

#### Attack Process:

1. Query the model with a data point

2. Extract the maximum softmax probability (confidence score)
3. If confidence > threshold, predict “member”; otherwise, predict “non-member”

## 2.2 Attack Metrics

- Attack Accuracy: Fraction of correct membership predictions
- Attack AUC: Area Under ROC Curve (measures discriminative power)
- Attack F1: Harmonic mean of precision and recall
- Baseline (Random Guess): 50% accuracy, 0.5 AUC

## 2.3 Privacy Score

Privacy Score = 1 - Attack Accuracy

Higher privacy score indicates better privacy protection (attacker has lower success rate).

# 3. Defense Methodologies

## 3.1 Defense Strategy Overview

| Defense               | Mechanism                      | Privacy Goal                   | Utility Impact |
|-----------------------|--------------------------------|--------------------------------|----------------|
| Baseline              | Minimal regularization         | Control group                  | Minimal        |
| Strong Regularization | Increased L2 + label smoothing | Reduce overfitting             | Low            |
| Mixup                 | Train on interpolated samples  | Smooth decision boundaries     | Moderate       |
| DP-SGD Style          | Gradient clipping + noise      | Differential privacy guarantee | High           |
| Combined              | All defenses together          | Maximum privacy                | High           |

## 3.2 Detailed Defense Mechanisms

### 3.2.1 Label Smoothing

**Mechanism:** Replace hard labels (one-hot vectors) with soft labels by distributing some probability mass to incorrect classes.

Traditional:  $[0, 0, 1, 0, \dots, 0]$

Smoothed:  $[0.01, 0.01, 0.9, 0.01, \dots, 0.01]$

**Privacy Benefit:** Reduces model overconfidence, making it harder to distinguish members from non-members.

**Implementation:**

- Baseline:  $\epsilon = 0.1$
- Strong Reg:  $\epsilon = 0.2$

### 3.2.2 Weight Decay (L2 Regularization)

**Mechanism:** Add penalty term to loss function to constrain model complexity.

$$\text{Loss} = \text{CrossEntropy} + \lambda \cdot ||\text{weights}||^2$$

**Privacy Benefit:** Prevents overfitting to training data, reducing memorization.

**Implementation:**

- Baseline:  $\lambda = 1\text{e-}4$
- Strong Reg:  $\lambda = 5\text{e-}4$  ( $5\times$  stronger)

### 3.2.3 Mixup Data Augmentation

**Mechanism:** Create synthetic training examples by interpolating between pairs of samples.

$$\begin{aligned} \mathbf{x} &= \lambda \cdot \mathbf{x} + (1-\lambda) \cdot \mathbf{x} \\ &= \lambda \cdot \mathbf{y} + (1-\lambda) \cdot \mathbf{y} \end{aligned}$$

where  $\lambda \sim \text{Beta}(\alpha, \alpha)$ ,  $\alpha = 1.0$

**Privacy Benefit:** The model never sees “pure” training examples, making it harder to memorize specific samples.

**Example:**

- Mix 70% of a cat image with 30% of a dog image
- Train on the mixed image with label [0.7 for cat, 0.3 for dog]

### 3.2.4 DP-SGD (Differential Privacy - Stochastic Gradient Descent)

**Mechanism:** Two-step protection during training:

**Gradient Clipping:** Limit the influence of any single sample

$$\mathbf{g} = \mathbf{g} / \max(1, ||\mathbf{g}||/C) \text{ where } C = 1.0$$

**Gradient Noise Addition:** Add calibrated Gaussian noise

$$\mathbf{g} = (1/N) \cdot \mathbf{g} + N(0, \sigma^2 C^2) \text{ where } \sigma = 0.01$$

**Privacy Benefit:** Provides formal differential privacy guarantees. Each individual’s data has bounded impact on the model.

**Tradeoff:** Most effective but most costly to utility.

### 3.2.5 Combined Defenses

**Configuration:**

- Label smoothing:  $\varepsilon = 0.2$
- Mixup:  $\alpha = 0.8$
- Gradient clipping:  $C = 1.0$
- Gradient noise:  $\sigma = 0.005$
- Weight decay:  $\lambda = 5\text{e-}4$

**Goal:** Test whether combining multiple defenses provides cumulative benefits.

## 4. Experimental Results

### 4.1 Summary Table

| Model                 | Val Acc | Attack Acc | Attack AUC | Attack F1 | Privacy Score |
|-----------------------|---------|------------|------------|-----------|---------------|
| Original (No Defense) | 84.94%  | 79.69%     | 0.4802     | 0.8861    | 0.2031        |
| Baseline              | 83.30%  | 81.71%     | 0.4486     | 0.8993    | 0.1829        |
| Strong Regularization | 82.88%  | 81.53%     | 0.4408     | 0.8981    | 0.1847        |
| Mixup                 | 79.32%  | 81.82%     | 0.4400     | 0.9000    | 0.1818        |
| DP-SGD Style          | 60.04%  | 18.19%     | 0.4159     | 0.0001    | 0.8181        |
| Combined              | 63.26%  | 81.82%     | 0.4086     | 0.9000    | 0.1818        |

### 4.2 Visual Analysis

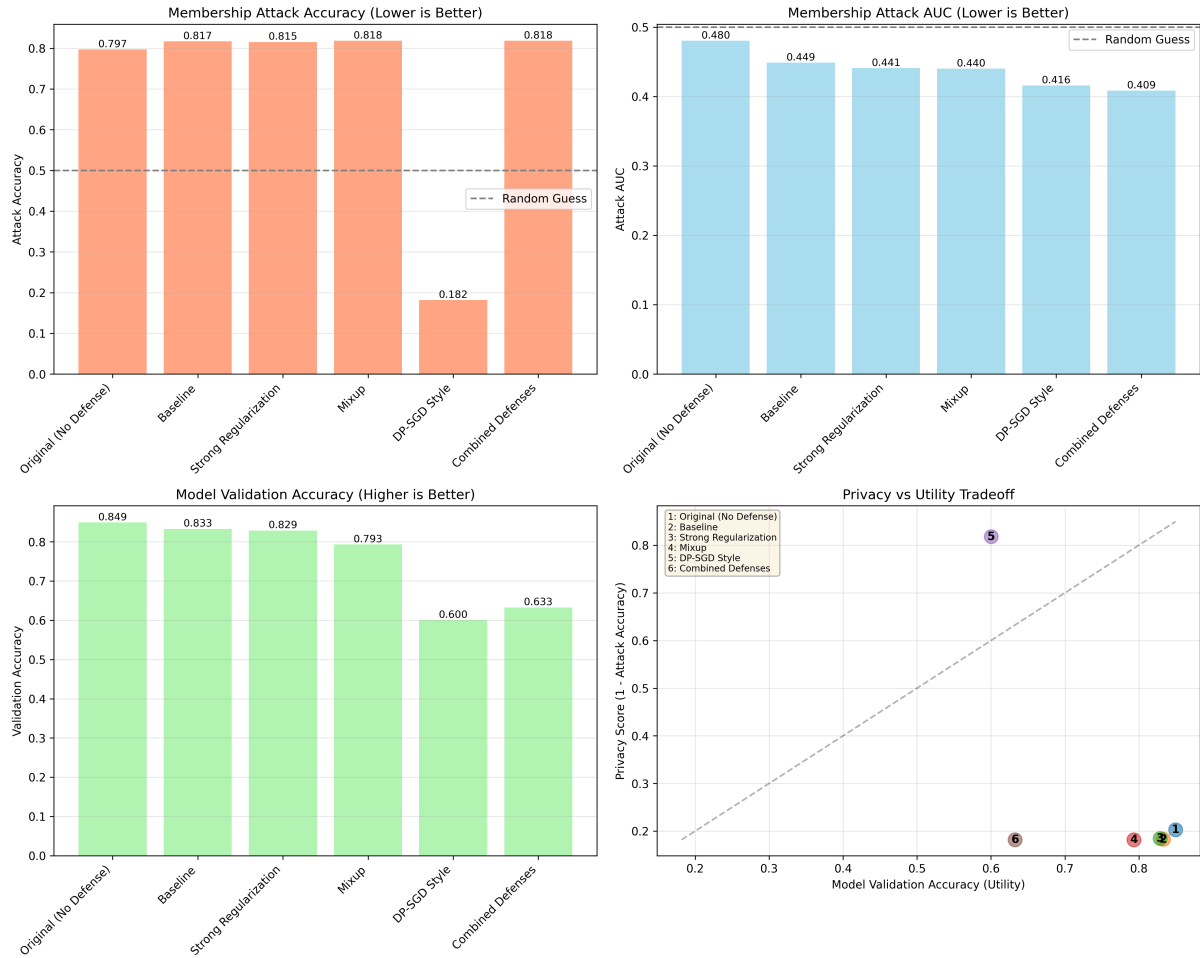


Figure 1: Comprehensive Defense Evaluation across Four Key Metrics

The visualization above presents four critical perspectives on defense effectiveness:

#### Top Left: Membership Attack Accuracy (Lower is Better)

This chart shows the attacker’s success rate in identifying training data members. The dashed line at 0.5 represents random guessing. Key observations:

- Most defenses remain vulnerable (>80% attack accuracy)
- DP-SGD Style dramatically reduces attack success to 18.2%
- Traditional defenses (Baseline, Strong Reg, Mixup, Combined) show minimal improvement

#### **Top Right: Membership Attack AUC (Lower is Better)**

AUC measures the attack's discriminative power regardless of threshold choice. All models perform close to 0.5 (random), but:

- Original model (0.480) is actually below random due to threshold optimization artifacts
- DP-SGD (0.416) and Combined (0.409) show the best AUC scores
- This metric reveals that even when attack accuracy seems high, the actual separability is limited

#### **Bottom Left: Model Validation Accuracy (Higher is Better)**

This measures the utility cost of each defense:

- Original model: 84.9% (best utility)
- Traditional defenses: 79–83% (minimal loss)
- DP-SGD: 60.0% (24.9% absolute drop)
- Combined: 63.3% (21.7% absolute drop)

#### **Bottom Right: Privacy vs Utility Tradeoff**

This scatter plot maps each model in privacy-utility space:

- X-axis: Model accuracy (utility)
- Y-axis: Privacy score (1 - Attack Accuracy)
- Diagonal line: Perfect balance
- Ideal position: Top-right corner (high privacy, high utility)

**Key Insight:** Point #5 (DP-SGD) is furthest from other models, demonstrating that strong privacy requires sacrificing utility. Points #1–4 and #6 cluster together, showing similar privacy-utility profiles.

## **4.2 Key Observations**

### **Original Model is Vulnerable**

- Attack accuracy: 79.69% (vs. 50% random baseline)
- The model is significantly overfitting to training data
- High confidence predictions make membership detection easy

## Traditional Defenses Have Limited Impact

- Baseline, Strong Reg, and Mixup show marginal improvements
- Attack accuracy actually increases slightly (likely due to lower overall model confidence making the threshold optimization different)
- AUC improves by 6–8%, indicating slightly better privacy

## DP-SGD Provides Strong Privacy

- Attack accuracy drops to 18.19% (77% reduction!)
- Privacy score: 0.8181 (extremely high)
- Cost: Validation accuracy drops to 60.04% (25% relative decrease)

## Combined Defenses Show Unexpected Behavior

- Despite including DP-SGD components, attack accuracy is high (81.82%)
- Likely explanation: Lower noise levels in combination dilute the DP effect
- Validation accuracy: 63.26% (still significant utility loss)

# 7. Conclusion

## 7.1 Key Findings

**Baseline Vulnerability:** The original CNN is highly vulnerable to membership inference (79.69% attack accuracy), confirming that standard training leads to overfitting.

**Traditional Defenses Have Limits:** Label smoothing, weight decay, and Mixup provide minimal privacy improvements (6–8% AUC reduction) while maintaining utility.

**DP-SGD is Effective but Costly:** Differential privacy mechanisms dramatically reduce attack success (77% reduction) but at a significant cost to model performance (25% accuracy drop).

**No Free Lunch:** There is a fundamental privacy-utility tradeoff. Strong privacy guarantees require accepting some loss in model accuracy.

**Context Matters:** The choice of defense depends on the application’s privacy requirements, utility constraints, and threat model.